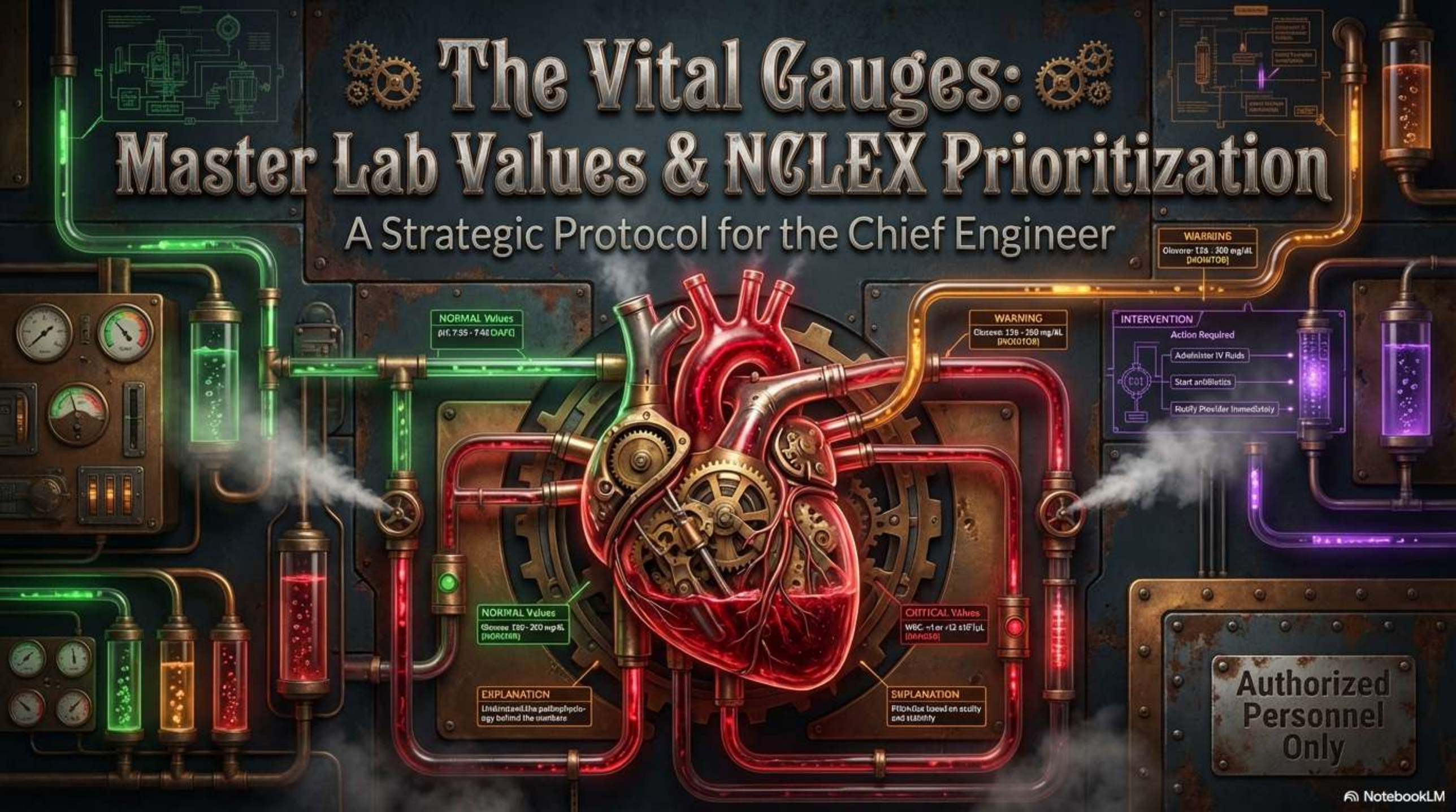
The Vital Gauges: Master Lab Values & NCLEX Prioritization

A Strategic Protocol for the Chief Engineer



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The Priority Protocol (The ABCD Framework)

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The Filter (Renal Function)

Exception:
Dye procedures
= Level B.

Creatinine
(Serum)

0.6 – 1.2 mg/dL

Level A: Best indicator of kidney function. Not an acute emergency.

BUN
(Blood Urea Nitrogen)

8 – 25 mg/dL

Level A/B: Indicates Dehydration.

BUN = Hamburger Buns come in packs of 8. (Range 8-25).

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Viscosity Control (Coagulation - INR)

Normal INR: **0.8 – 1.2**

Therapeutic (Warfarin): **2 – 3**

CRITICAL SCENARIO (Level C):

INR > 4

Action Protocol

1. **HOLD** (Warfarin)
2. **ASSESS** (Bleeding)
3. **PREPARE** (Vitamin K)
4. **CALL** (Provider)

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DANGER: HIGH VOLTAGE

The Energy Source (Potassium - K^+)

Normal Range: 3.5 – 5.3 mEq/L

Low (< 3.5) = Level C

Action: Assess Heart -> Prepare K+ -> Call.

High (5.4 – 5.9) = Level C

Action: Hold K+ -> Assess Heart -> Prepare Kayexalate/D5W/Insulin -> Call.

High (> 6.0) = Level D (DEADLY)

Action: STAT. Do Not Leave Bedside. Team approach required.

ALARM

Mnemonic: "Kalium" pumps the heart.

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The Pressure Valves (pH Balance)

Critical < 7

pH: 7.35 – 7.45

HCO_3 (Bicarb): 22 – 26

$PaCO_2$: 35 – 45

CRITICAL ALERT (Level D):
pH in the 6s (e.g., 6.8).
The Engine is Shutting Down.

Action Protocol

1. Assess Vitals (Check for life)
2. Call Provider Immediately
3. STAY AT BEDSIDE

Mnemonic: R.O.M.E.
(Respiratory Opposite, Metabolic Equal)

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RESPIRATORY EXHAUST (CO_2)

Normal CO_2: 35 – 45 mmHg

High (50s) – Level C

Pathology: Retaining CO2 (Non-COPD).
Action: Pursed Lip Breathing.

High (60s) = Respiratory Failure (Level D)

Action Protocol: Prepare to Intubate/Ventilate -> Call Respiratory -> Call MD. DO NOT LEAVE ROOM.

ALARM

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Oxygenation (Intake Mixture)

SaO_2 (Saturation): 98%

pO2 (Arterial): 78 – 100 mmHg
SaO2 (Saturation): 93 – 100%

Prioritized

Level C: pO2 70-77
-> **Action:** Give Oxygen.

Level D (Respiratory Failure):
pO2 in the 60s -> **Action:** Intubate.

Respiratory Failure Criteria:
CO2 > 60 AND pO2 < 60.

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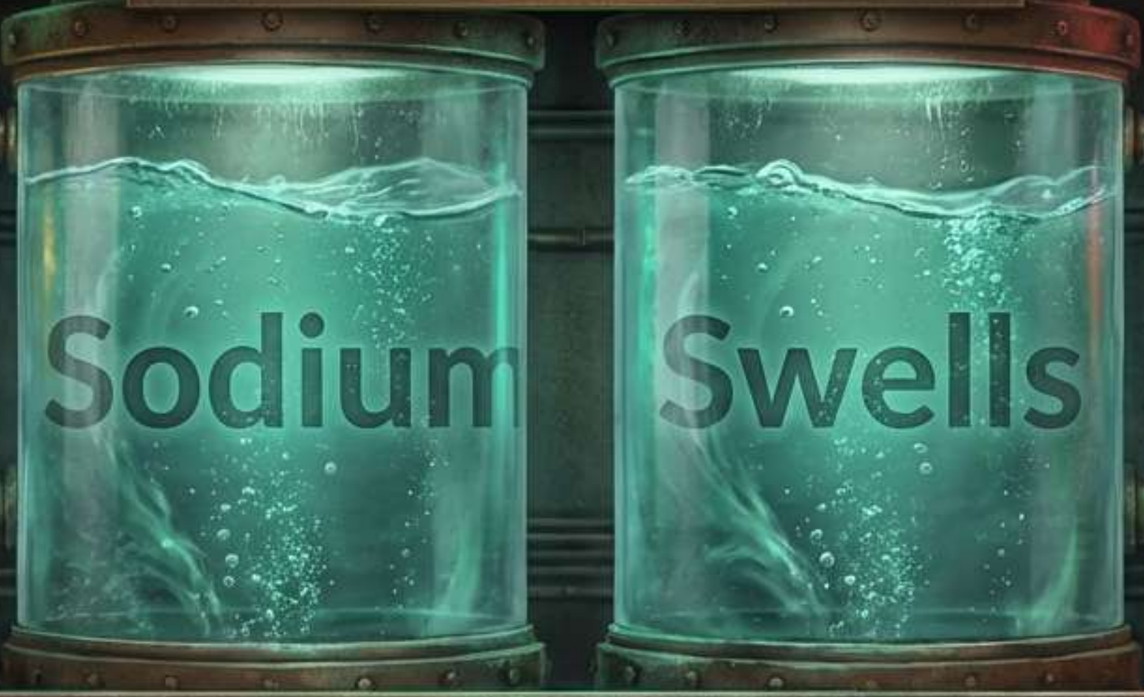
Fluid Dynamics (Sodium & Electrolytes)



WARNING (Level B)

Abnormal Na+ -> Assess Dehydration or Overload.

CRITICAL (Level C)

Abnormal Na+ WITH Change in LOC -> Safety Issue.

Sodium (Na^+): 135 – 145 mEq/L

Magnesium (1.3 – 2.1) = 'Mellows Muscles' (Low = Twitching)
Calcium (9.0 – 10.5) = "Contracts Muscles" (Low = Tetany)

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Hematology (Fuel Quality)

Hemoglobin (Hgb): 12 – 18 g/dL

Hematocrit (Hct): 36 – 54% (3x Hgb)

Hgb 8 — 18
Action: Asses bleeding, Prepare Transfusion, & Call.

Level B: Hgb 8 – 11 -> **Action:** Assess bleeding/anemia/malnutrition.

Level C: Hgb < 8 -> **Action:** Assess bleeding, Prepare Transfusion, Call.

Hgb x 3 = Hct

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Infection Control (WBCs)

Total WBC: 5,000 – 11,000
ANC (Absolute Neutrophil Count): > 500
CD4: > 200

CRITICAL (Level C)
Critical if:
WBC < 5,000 **OR**
ANC < 500 **OR** CD4 < 200

NEUTROPENIC PRECAUTIONS:
- Strict handwashing
- No fresh flowers or potted plants
- No standing water (pitchers)
- Avoid crowds

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Platelets (Hull Integrity)

Platelets:
150,000 – 400,000

Level C: < 90,000
-> Action: Bleeding Precautions (Soft toothbrush, No IM injections).

Level D (DEADLY): < 40,000
-> Action: High risk of spontaneous hemorrhage.

Value	Abbreviation	Unit	Normal Range
Prothrombin Time	PT	Seconds	11 - 14
Partial Thromboplastin Time	PTT	Seconds	25 - 35
International Normalized Ratio	INR		0.8 - 1.2
Activated Partial Thromboplastin Time	aPTT		1.5 - 2.5

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Engine Diagnostics (Cardiac Labs)

Troponin > 0.5 = MI (Infarction).
Priority: Acute/Critical.

Troponin > 0.5 = MI (Infarction).
Priority: Acute/Critical.

BNP > 100 = CHF (Failure).
Priority: Level B (Chronic Monitor).

Strategy: Do not prioritize a high BNP (Chronic) over an acute lab like Potassium or Troponin.

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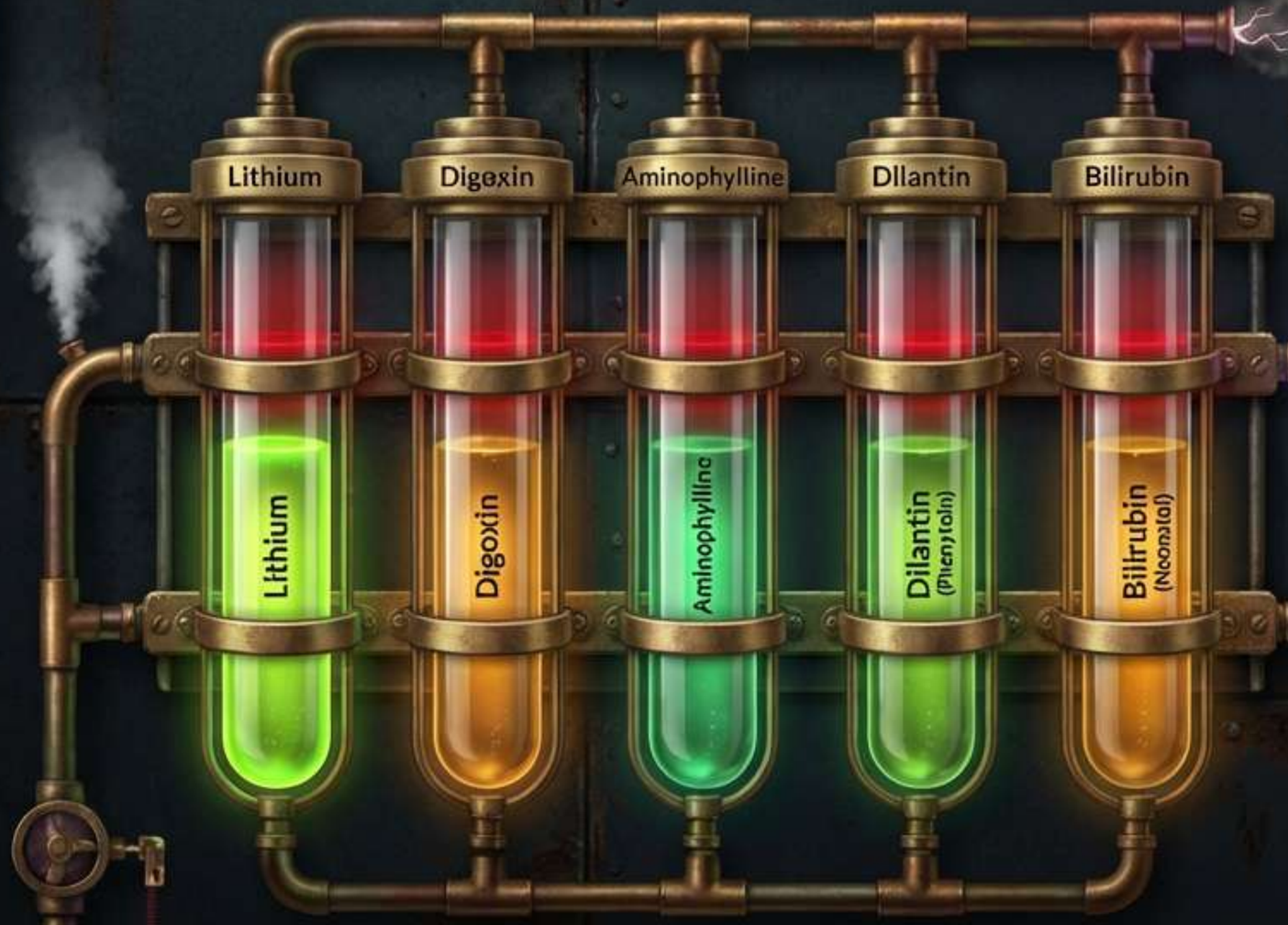
THE THERAPEUTIC INDEX (PHARMACOLOGY)



Lithium:	0.6 – 1.2 (Toxic > 2)
Digoxin:	0.5 – 2 (Toxic > 2)
Aminophylline:	10 – 20 (Toxic > 20)
Dilantin (Phenytoin):	10 – 20 (Toxic > 20)
Bilirubin (Neonatal):	10 – 20 (Toxic > 20)

The Magic 20s: Aminophylline, Dilantin, Bilirubin

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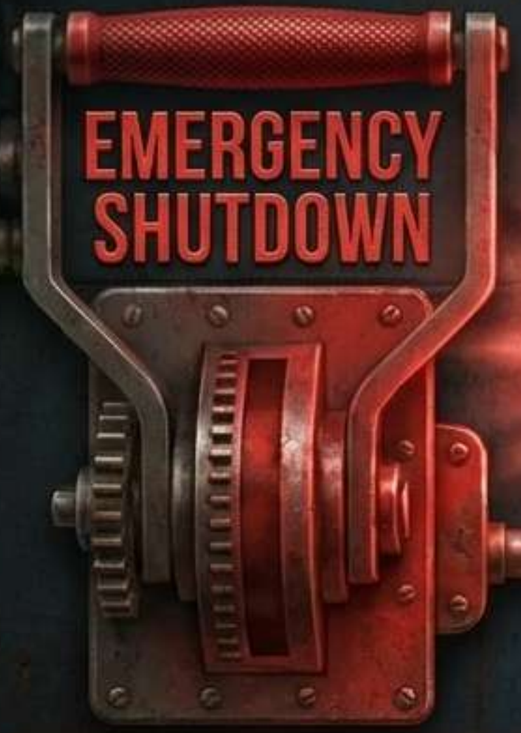
The Master ‘Level D’ Checklist



1. pH in the 6s
2. Potassium (K+) in the 6s
3. CO_2 in the 60s
4. pO_2 in the 60s
5. Platelets < 40,000

DO NOT LEAVE THE BEDSIDE.

The Chief Engineer maintains the watch.

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